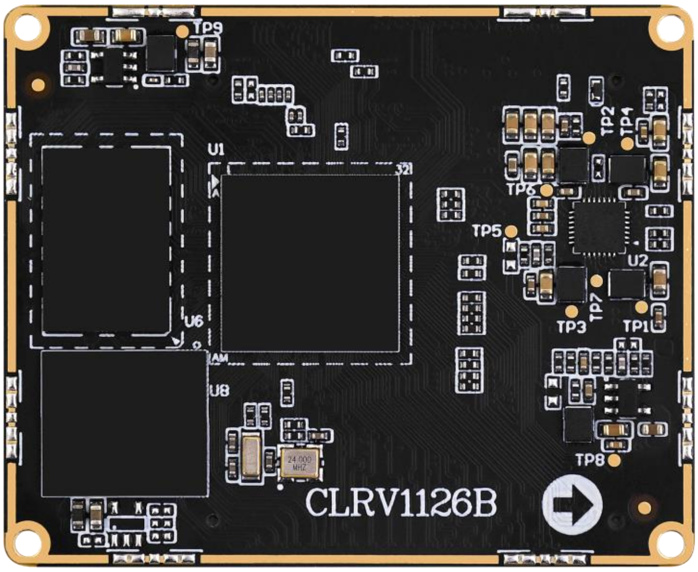
ATK-CLRV1126B

Core Board Specification

V1.0



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Disclaimer

In order to get the latest version of product information, please regularly visit the download center or contact the customer service of Taobao ALIENTEK flagship store. Thank you for your tolerance and support.

Revision History:

Version	Version Update Notes	Responsible person	Proofreading	Date
V1.0	release officially	ALIENTEK Linux Team	ALIENTEK Linux Team	2025.12

Catalogue

Chapter 1. Core board overview

1.1 Core board introduction

The Core Board RV1126B development board is based on the domestic Rockchip RV1126B/BJ processor, equipped with a powerful quad-core chip: 4×Cortex-A53 (up to 1.6 GHz), with an NPU providing 3 TOPS of computing power. It supports a 2D Graphics Engine (RGA), 4K hardware codec, and ISP.

The core board measures only 45 mm × 55 mm, and the matching baseboard is 140 mm × 180 mm. Commercial-grade or industrial-grade core boards are available with multiple configurations. It adopts a separate core-board plus baseboard design, with the core board connected to the baseboard via a BTB connector. The baseboard integrates serial ports, USB interfaces, Wi-Fi, Bluetooth, audio, Gigabit Ethernet, MIPI CSI×2, MIPI DSI, 4G/5G module interfaces, CAN FD, RS485, RTC, and more.

It boasts powerful edge AI computing capabilities and supports mixed-precision computing of INT8/INT16. It covers core visual task networks such as object detection, image segmentation, and image classification. Meanwhile, it integrates advanced speech processing technologies, including highly natural text-to-speech (TTS) and accurate automatic speech recognition (ASR), and is equipped with a full-process OCR text detection and recognition solution. The system is deeply compatible with mainstream deep learning frameworks such as PyTorch, TensorFlow, and ONNX.

It supports hard decoding of H.265/H.264 up to 4K (3840×2160) @30fps. It supports HEVC (H.265) Main Profile Level 5.0 High Tier, H.264 High Profile Level 5.0, and JPEG Baseline up to 12M@30fps. It also supports hard encoding of mainstream resolutions including 4K (3840×2160) @30fps and 1080P@30fps. It can be applied in intelligent security, smart door locks, industrial vision, smart home, intelligent vehicle, robotics, and other fields.

The core board leads out a total of 116 GPIO pins (which can be multiplexed for other functions) and 60 functional pins (for MIPI camera, MIPI display, USB, ADC, etc.).

The core board is equipped with comprehensive development documents and software resources, all of which are freely available. To improve user development efficiency and shorten development cycles, ALIENTEK has specially compiled a full set of materials for core board users, including schematics, baseboard design files, mechanical structures, component footprints, connector specifications, factory system image source code, compilers, software packages, and more, to facilitate user development.

Enterprise customers purchasing core boards in bulk can contact the Linux technical support team at the official ALIENTEK store on Taobao, so that a dedicated WeChat Work technical support group can be established exclusively for enterprise customers. Through this dedicated channel, we are committed to providing customers with more efficient and professional technical support, ensuring prompt and accurate responses and assistance during use.

1.2 Purchase Channels

ALIENTEK Official Store:
https://zhengdianyuanzi.tmall.com
ALIENTEK Data Download Center:
http://www.openedv.com/docs/boards/arm-linux/index.html

Chapter 2. Core board hardware parameters

2.1 Hardware Parameters

Item	Parameter	Note
Size	55mm*45.00mm	Length * Width
CPU	RV1126B	BGA491 package
Memory	1GB/2GB LPDDR4/4X	Surface mount packaging. Due to the impact of chip supply, there may be chips from various manufacturers. All will be based on the actual model used for assembly.
Storage	8GB/32GB EMMC	Surface mount packaging. Due to the impact of chip supply, there may be chips from various manufacturers. All will be based on the actual model used for assembly.
Operating voltage	5.0V	
Power consumption	>1.8W	Static power consumption. The specific power consumption depends on the peripheral.
Operating temperature	Commercial grade: 0℃ ~ +70℃ Industrial grade: -40℃ ~ +85℃	RV1126B is commercial grade, while RV1126BJ is industrial grade; the two are compatible.
Number of pins	240Pin	
Pin spacing	0.5mm	
Core board connection method	BTB	
PCB process	8 layers, gold plating process, independent grounding signal layer	Adopt lead-free technology

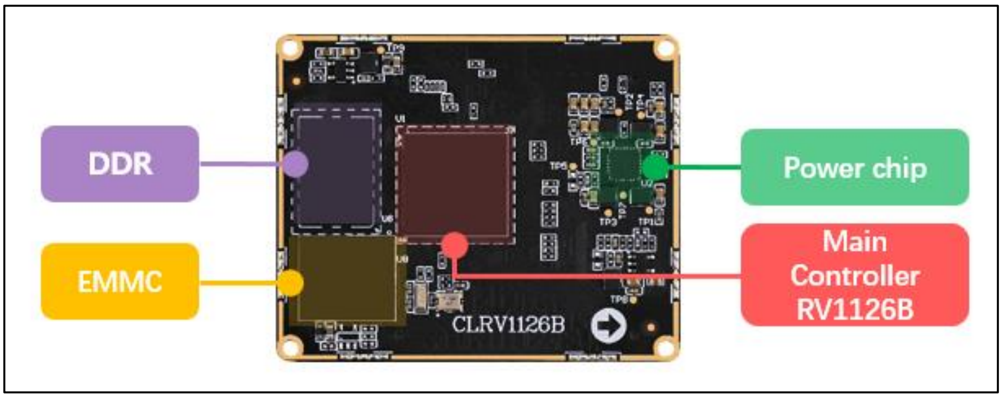

Figure 2.1-1 Front resources of the core board

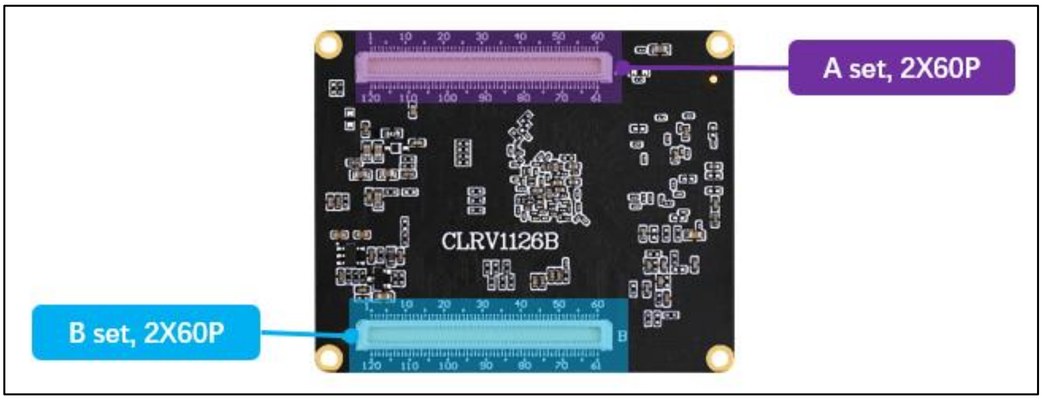

Figure 2.1-2 Core Board Pins

2.2 Parameters of RV1126B Chip

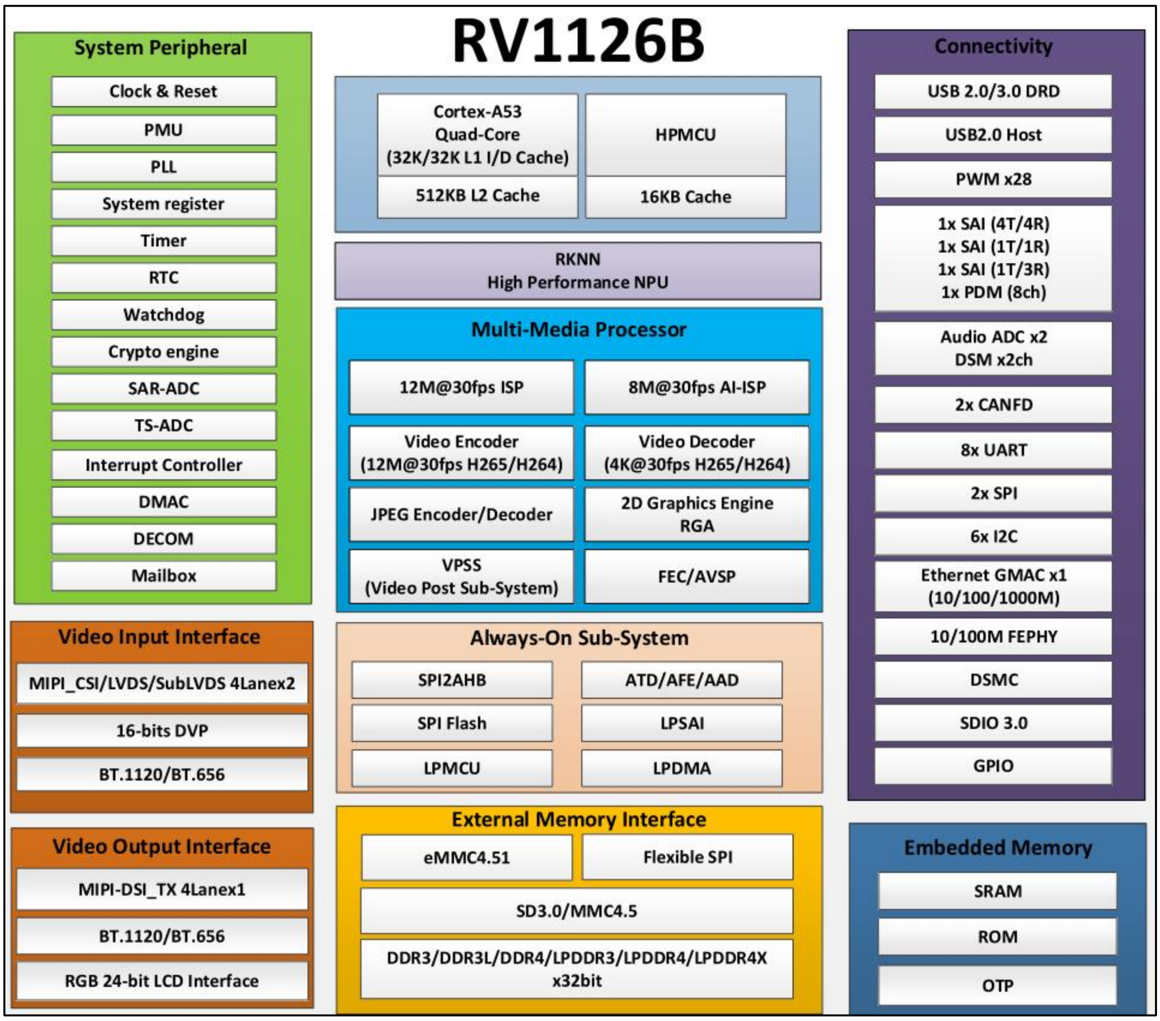

Figure 2-2.1 Chip resources

For specific detailed parameters, please refer to the data sheet of the RK3568 chip.

RV1126B Main Controller Chip Resources	
Processor	Quad-core ARM Cortex-A53
NPU	3.0 TOPS (supports INT8/INT16 mixed-precision computing)
ISP	Supports HDR, 3A, LSC, 3DNR, 2DNR Maximum input: 12M@30fps
RGA	Input formats: ARGB8888, RGBA8888, RGB888, RGB565, YUV422/420/444 (Planar/Semi-planar, 8/10 bit) and other mainstream formats. Output formats: ARGB8888, RGBA8888, RGB888, RGB565, YUV420/422/444 (Planar/Semi-planar), YUV400. Maximum resolution: 8192×8192 (source), 4096×4096 (destination)
Video Decoder	H.264、H.265：3840x2160@30fps
Video Encoder	H.265, H.264: Supports hardware encoding for mainstream resolutions including 4K (3840×2160)@30fps and 1080P@30fps
JPEG Encoder	JPEG Baseline encoding Supports parallel encoding (e.g., HEVC+JPEG or H.264+JPEG) Maximum support: 12M@30fps
JPEG Decoder	Supports YUV400/YUV420/YUV422/YUV440/YUV411/YUV444 Supported image size range: 48×48 to 65520×65520
Display Output Interface	MCU/RGB LCD interface: up to 24 bits BT.656/BT.1120 interface 4-lane MIPI interface: 1.5 Gbps/lane, supports up to 1920×1080@60fps
Video Input Interface	2×MIPI CSI, each with MIPI D-PHY V1.2, 2 lanes, up to 2.5 Gbps per lane
Audio Interface	3×SAI: SAI0: 4 TX and 4 RX SAI1: 1 TX and 1 RX SAI2: 1 TX and 3 RX
Ethernet Interface	One Ethernet port: Integrated Ethernet PHY supporting 10/100 Mbps compliant with IEEE 802.3/802.3u. Supports 10/100/1000 Mbps via RGMII interface.
USB Controller	One USB 2.0 Host: supports High-Speed (480 Mbps), Full-Speed (12 Mbps), and Low-Speed (1.5 Mbps) modes. One USB 3.0
SPI	2×SPI
I2C	6×I2C: supports 7-bit and 10-bit addressing modes; I2C data rates: Standard mode: up to 100 kbps Fast mode: up to 400 kbps

	High-speed mode: up to 1 Mbps
UART	8× [UART]
PWM	4× [PWM]
CAN	2×: supports CAN and CAN FD
ADC	3× SAR ADC (13-bit resolution, sampling rate up to 2 MS/s)

Note: These are the resource parameters of the chip datasheet, not the available resource parameters of the core board.

2.3 Reusable Resources of Core Board Pins

The core board brings out all IOs from the processor. Users can design their own baseboard according to their needs to utilize the IO resources on the core board and configure the IOs to the required functions through pin multiplexing.

The default factory firmware of the core board only supports the functions described in Section 2.3 and cannot be directly used for other multiplexed functions. Firmware for multiplexed functions requires additional development.The core board pin multiplexing table can be found in the documentation disk.

Based on peripheral functions, the maximum number of single peripherals that can be multiplexed on the ATK-CLRV1126B core board is listed here. Specific configuration can refer to the chip datasheet.

The following references are from the RV1126B chip datasheet.(Maximum number of single peripherals: refers to the maximum quantity of a certain peripheral that the core board can use under the premise that no other peripherals are occupied.)

Pin peripheral function	Maximum	Note
MIPI Display Interface	1	4-lane MIPI DSI, supporting up to1920×1080@60fps
MIPI Camera Interface	2	4-lane MIPI CSI, with a transmission rate of 2.5 Gbps per lane
PWM	4	10/100 M Ethernet controller Alternatively supports 10/100/1000 Mbps via RGMII interface
SAI	3	
Ethernet	1	
USB 2.0	1	Can be expanded via USB hub
USB3.0	1	Can be expanded via USB hub
SPI	2	Supports master mode and serial slave mode
I2C	6	Supports 7-bit and 10-bit address modes
UART	8	Built-in 8 UARTs; UART0 is used for core board debug serial port
CAN	2	
ADC	3	Built-in 3 SAR ADCs

*Note: The ATK-DLRV1126B Development Board and ATK-CLRV1126B Core Board only support modules and accessories sold in the official ALIENTEK store. Other third-party modules require user-defined development or discussion and learning in the user group. Currently, all materials provided by ALIENTEK are available in the cloud disk at the official ALIENTEK Download Center.

Chapter 3. Core board software resources

3.1 Factory system software resources

The factory Linux system software resources are shown in Table 3.1 below:

Type	Description	Note
U-Boot	Version: 2017.09	Provide the data package.
Linux Kernel	Version: 6.1	Provide the data package.
Buildroot	Verson: 2024.02	Provide the data package.
Qt5	Version: 5.15.11	Provide the data package.
Cross compiler	aarch64-buildroot-linux-gnu-	Provide the data package.
System Flashing Method	Upper computer programming	Provides usage tutorials
MIPI LCD drive	MIPI DSI drive	Provide the data package.
Touch	GT911 capacitive touchscreen (available only from ALIENTEK)	Provide the data package.
Network	Gigabit Ethernet PHY is YT8531C	Provide the data package.
USB HOST	One USB HOST 2.0 One USB HOST 3.0	Provide the data package.
RESET button	Reset function	Provide the data package.
TF card	SDMMC drive	Provide the data package.
LED	GPIO	Provide the data package.
Audio	ES8390	Provide the data package.
USB WIFI&BT	RTL8733	Provide the data package.
Serial port	USB debugging serial port	Provide the data package.
ADC	ADC driver	Provide the data package.
PWM	LCD PWM backlight	Provide the data package.
CAN	2 CAN channels	Provide the data package.

Table 3.1.1 Linux System Software Resources Preinstalled on the Development Board

Chapter 4. Core Board Structural Dimensions

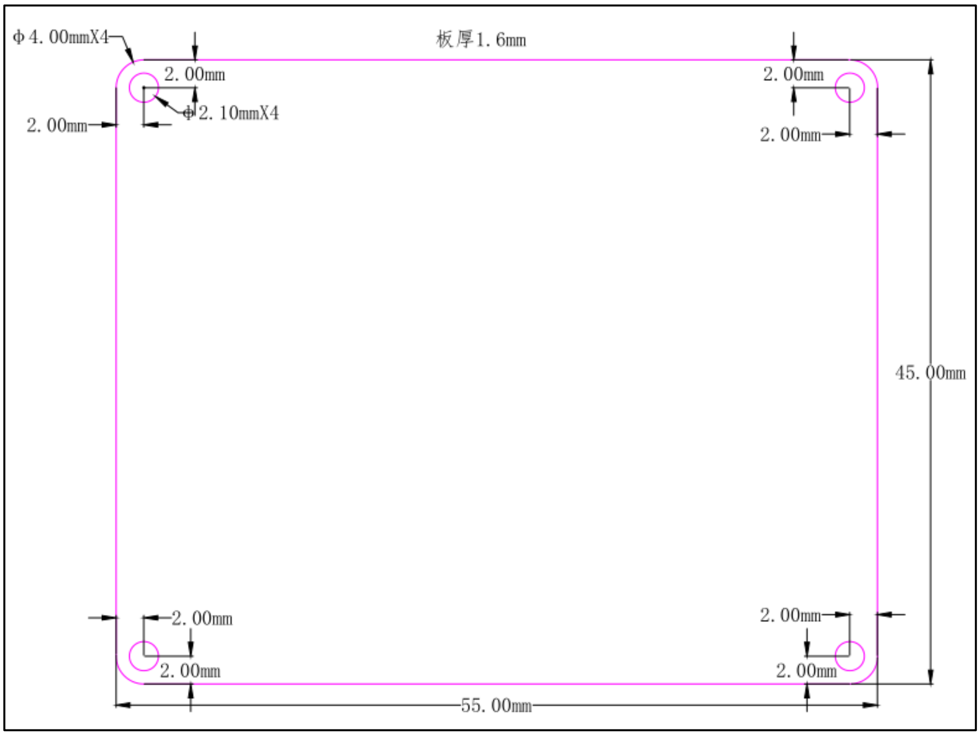

Figure 4-3.1-1 Core board structure size diagram

Chapter 5. Development materials

Download development materials:

http://www.openedv.com/docs/boards/arm-linux/index.html

Note: The development board documentation is continuously updated according to customer requirements. The cloud disk documentation shall prevail as the final official version.

The development materials are written based on the ATK-DLRV1126B Development Board. Please use the development board for project evaluation and testing.

Chapter 6. Optional accessories

6.1 ATK-DLRV1126B Development board

The following accessories can all be purchased at the ALIENTEK store.

https://zhengdianyuanzi.tmall.com

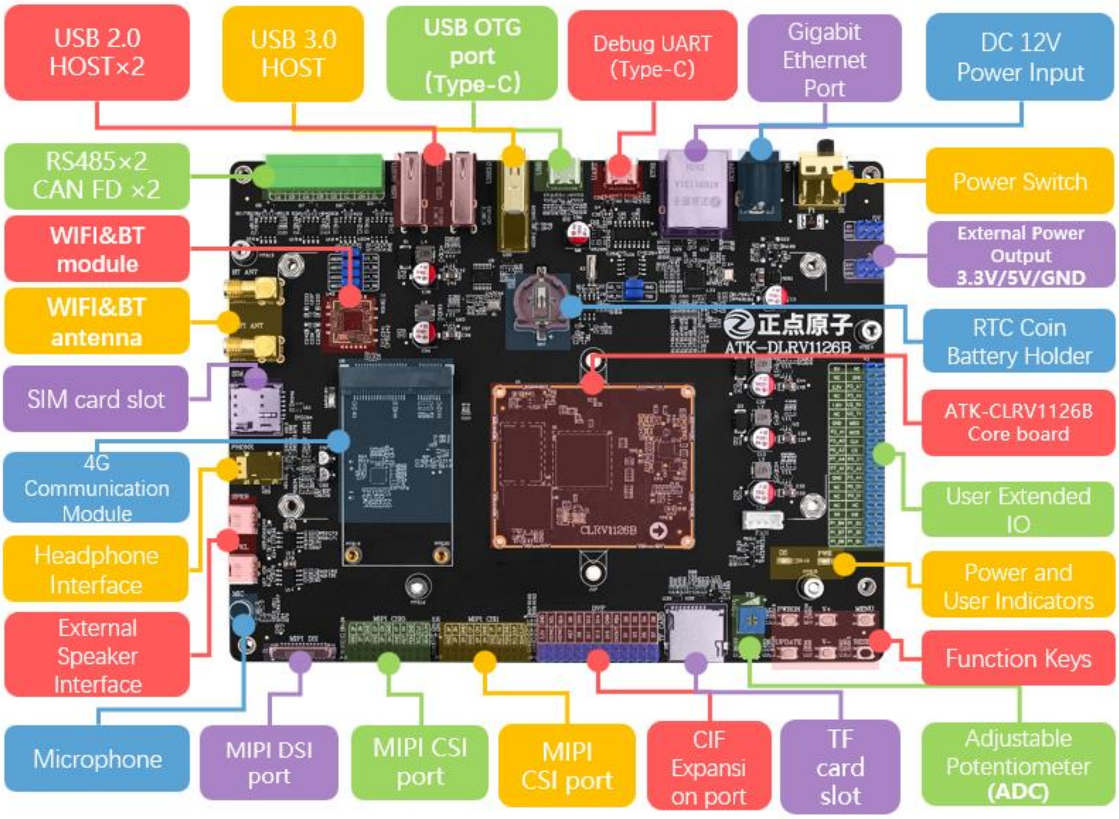

Figure 6.1-1 Front view of the ATK-DLRV1126B Development Board

Chapter 7. Precautions and maintenance

Notes

- Do not plug and unplug peripheral modules with power!
- Before using the product, please carefully read this manual and related development manuals, and pay attention to the applicable matters of the platform.
- Follow all instructions and warnings on the product.
- Please use this product in a cool, dry and clean place.
- Please keep the product dry. If any liquid splashes or soaks, power off immediately and let dry thoroughly.
- Do not use organic solvents or corrosive liquids to clean the product.
- Do not use or store this product in dusty, dirty and messy environment.
- If not used for a long time, please package this product, pay attention to moisture-proof and dust-proof.
- Pay attention to the ventilation and heat dissipation of the product during use to avoid component damage caused by excessive temperature during operation.
- Do not use this product in alternating hot and cold environment to avoid dew damage to components.
- Do not treat this product roughly, drop, knock or shake violently may damage the line and components.
- Pay attention to anti-static when using this product.
- FPC flexible cable is fragile, when plugging cable, pay attention to check whether the metal at both ends of the cable is misplaced and falling off.
- All products have passed the product test before shipment. Please use the development board corresponding to the ALIENTEK for power on test for the first time.
- Do not repair or disassemble the company's products by yourself. If the product fails, please contact the company in time for maintenance.
- Unauthorized modification or use of unauthorized parts may damage the product, the resulting damage will not be repaired.

Chapter 8. After sales service

8.1 Terms of after-sales service

1). After receiving the goods, please open them in front of the express, and sign after acceptance. If you find that the goods are less after signing, take photos in time and contact the seller's customer service to explain the situation within 15 days. If the feedback is lack of goods after 15 days, we will not reissue the goods. Other reasons notwithstanding).

2). 15 days -1 month: we are responsible for the return freight repair of product problems. Human factors damage expensive main chip or LCD screen, touch screen. The buyer needs to pay the cost and one time shipping fee, no maintenance fee.

3). 1-3 months: the problem of the product itself (non-human factors), we are responsible for the delivery of the past freight maintenance. If the main chip is burned out and the LCD screen and touch screen are damaged, the buyer needs to pay the cost, and the maintenance fee is not charged.

4) After 3 months: the buyer shall bear the return freight and the cost of chip, LCD screen and touch screen. No service charge.

8.2 After-sales Support

Technical support:

QQ group: ALIENTEK - Rockchip Communication group: 775783660

ALIENTEK – RV1126B User Group (order number required): 571769456

Taobao shop: ALIENTEK flagship store

Forum: http://www.openedv.com/forum-277-1.html